2026 Semester 1

Lecture 2 Supplement 1

01211373 Machine Learning and Programming for Industry

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Logistic regression

AI 's explanation on the term "regression"

- "The term "regression" in statistics comes from Sir Francis Galton's observations about heredity, specifically the tendency of offspring to regress towards the mean of a population. In the late 1800s, Galton studied the relationship between parents' and children's heights and noticed that exceptionally tall or short parents tended to have children whose heights were closer to the average height of the population. This "regression towards the mean" is the origin of the term, which later evolved into the broader statistical concept of regression analysis."

Logistic regression

- A supervised learning classification
- For binary classification , target y akes on two values {0,1}
- Hypothesis (or model)

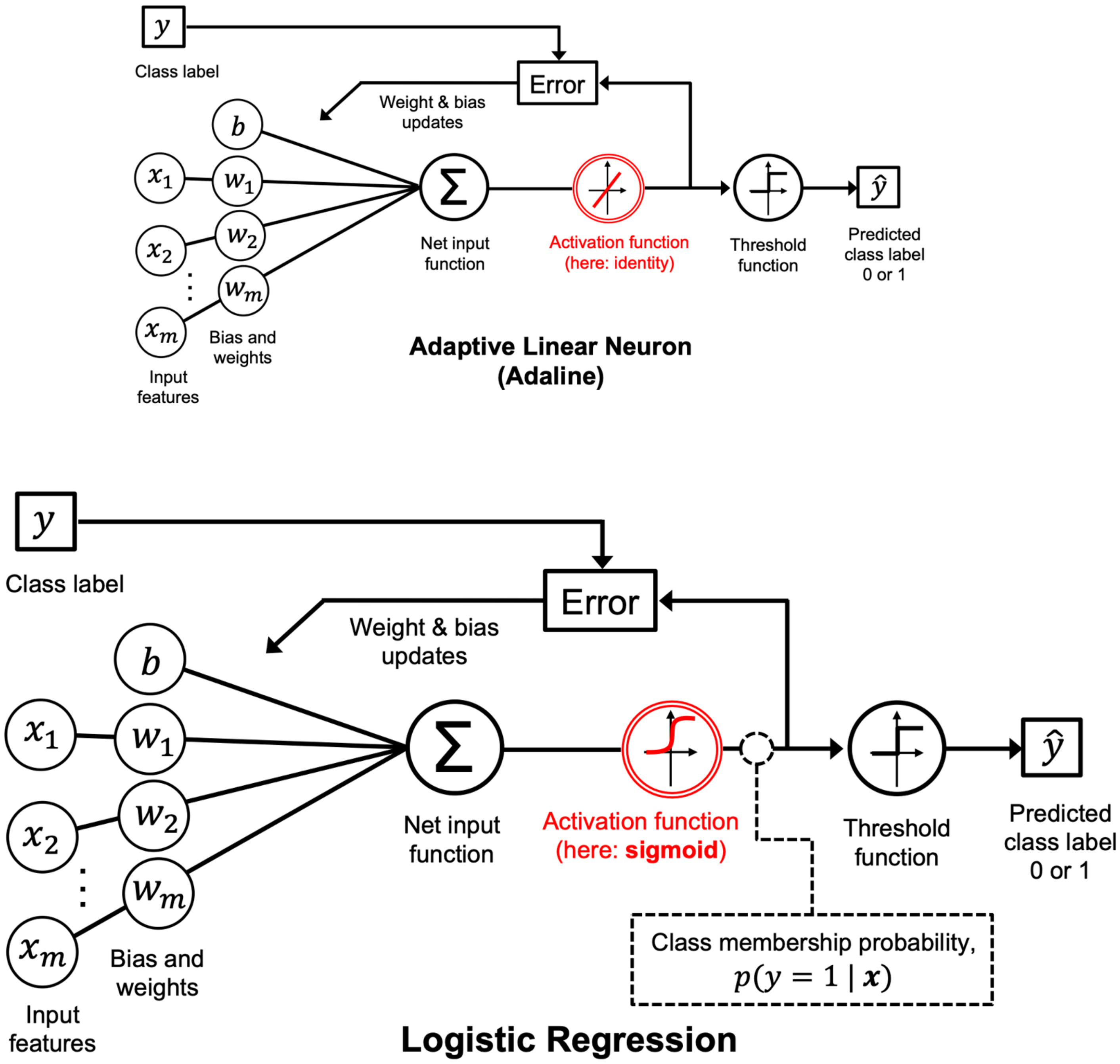
$$h_\theta(x) = g(\theta^T x) = \frac{1}{1 + e^{-\theta^T x}}$$

Read chapter 3 (p 59) of [1]

$$g(z) = \frac{1}{1 + e^{z}}$$

Called
Sigmoid function

Logistic regression compared to Adaline [1]



Sigmoid function

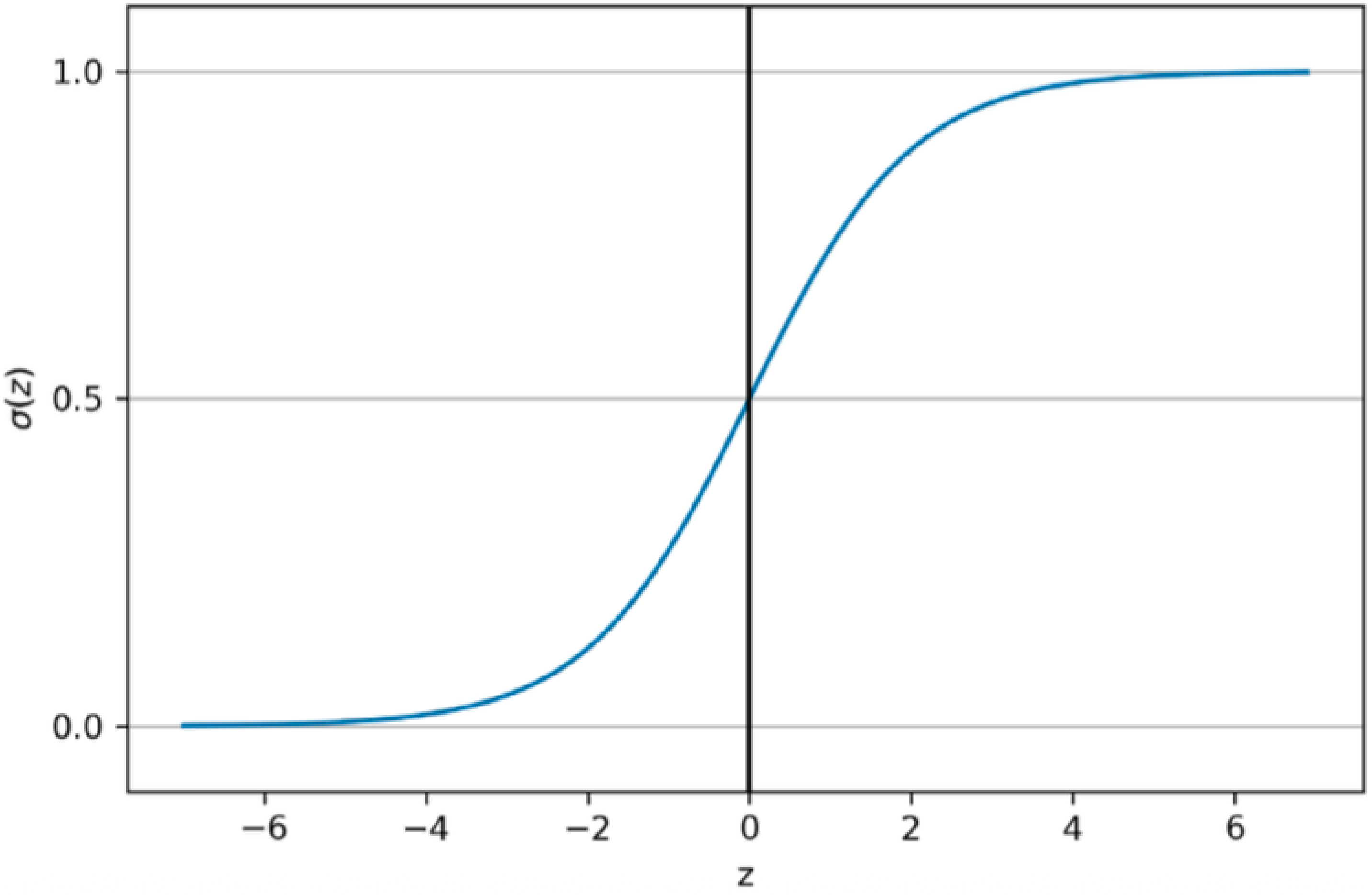
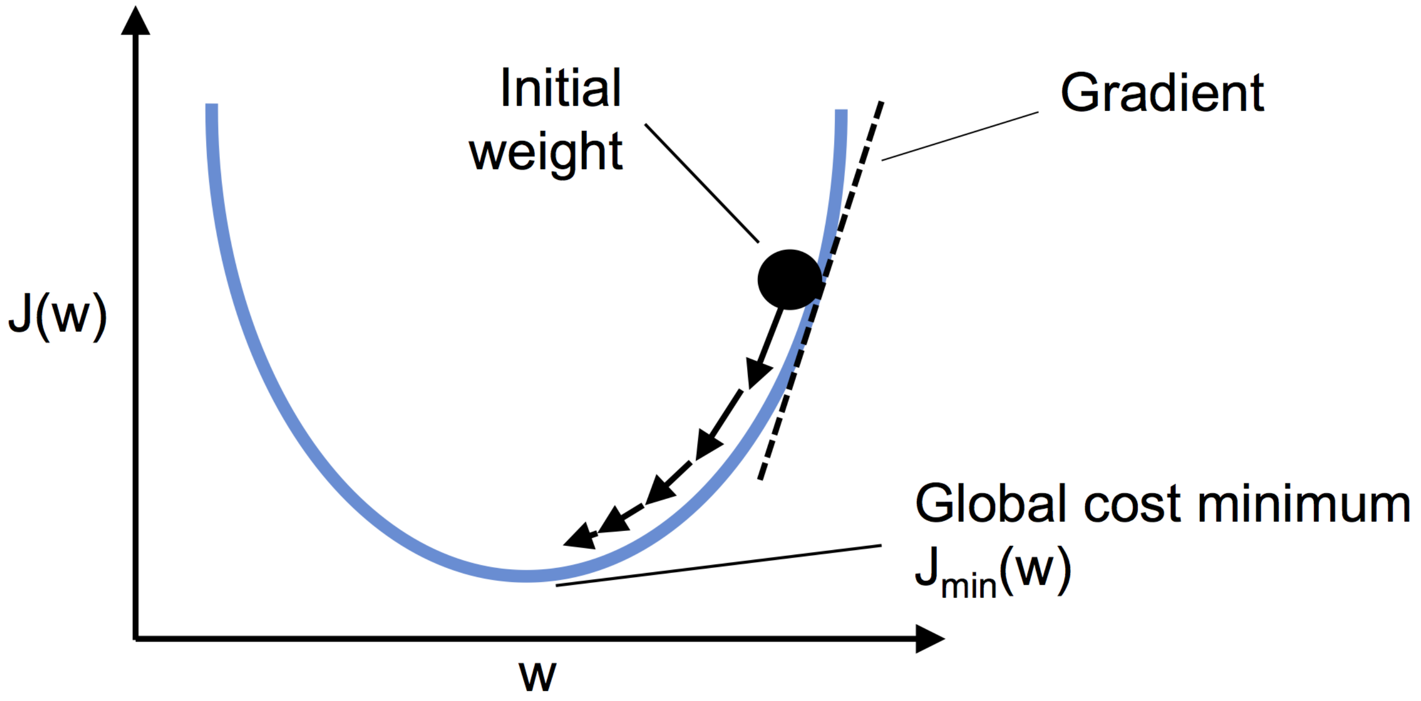


Figure 3.2: A plot of the logistic sigmoid function

use gradient descent to minimize loss function



calculating the partial derivative of the log-likelihood function

$$\frac{\partial L}{\partial w_j} = \underbrace{\frac{\partial L}{\partial a}\frac{da}{dz}\frac{\partial z}{\partial w_j}}_{\text{Apply chain rule}}$$

where $a = \sigma(z) = \dfrac{1}{1+e^{-z}}$

1) Derive terms separately:

$$\frac{\partial L}{\partial a} = \frac{a-y}{a-a^2}$$

$$\frac{da}{dz} = \frac{e^{-z}}{(1+e^{-z})^2} = a \cdot (1-a)$$

$$\frac{\partial z}{w_j} = x_j$$

2) Combine via chain rule and simplify:

$$\frac{\partial L}{\partial z} = a - y$$

$$\frac{\partial L}{\partial w_j} = (a-y)x_j$$
$$= -(y-a)x_j$$

-(y-a)

$$\Delta w_j = -\eta \frac{\partial L}{\partial w_j} \text{ and } \Delta b = -\eta \frac{\partial L}{\partial b}$$

weight and bias update

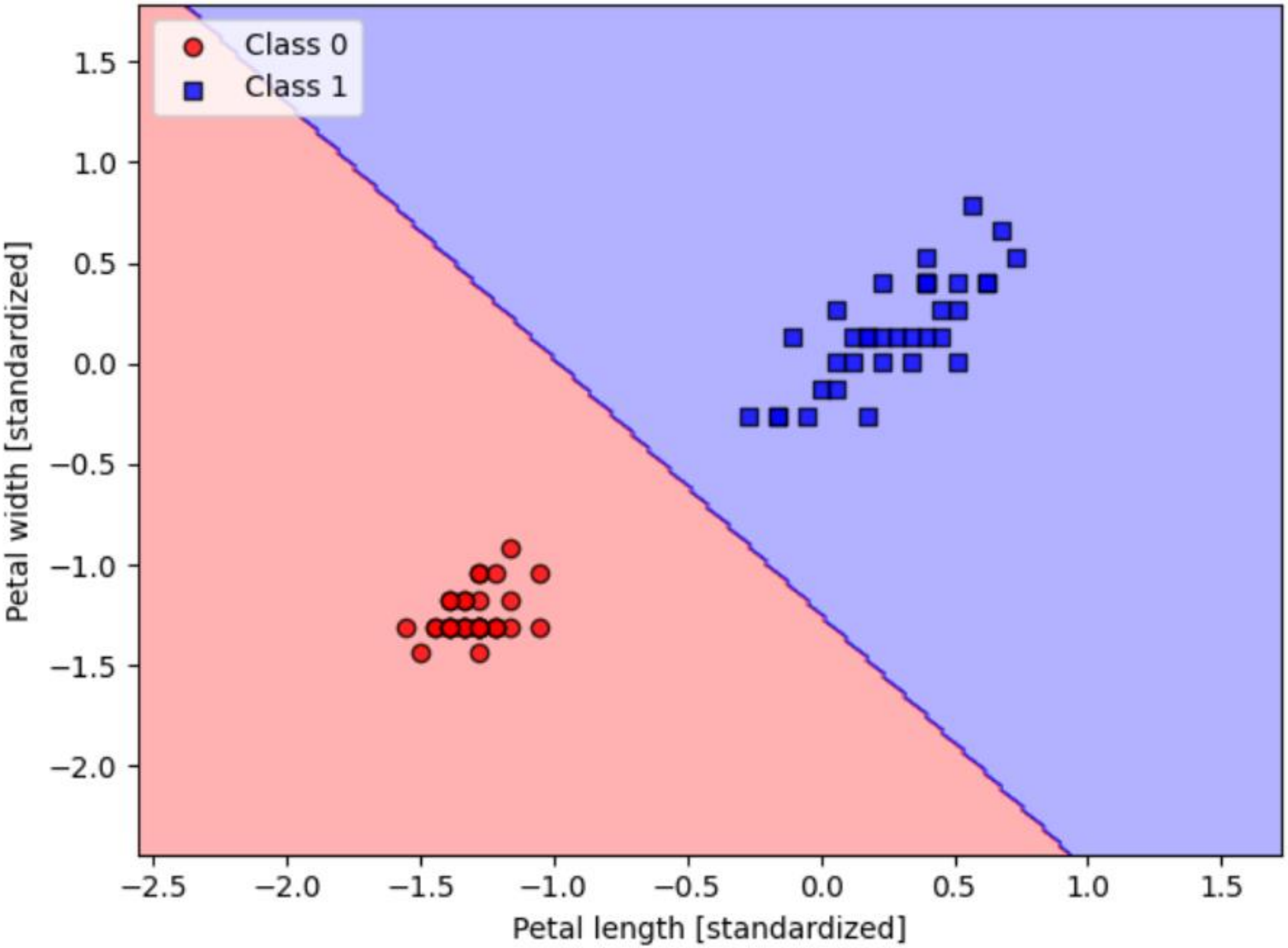
$$\boldsymbol{w} := \boldsymbol{w} + \Delta \boldsymbol{w}, \quad b := b + \Delta b$$

learning rate

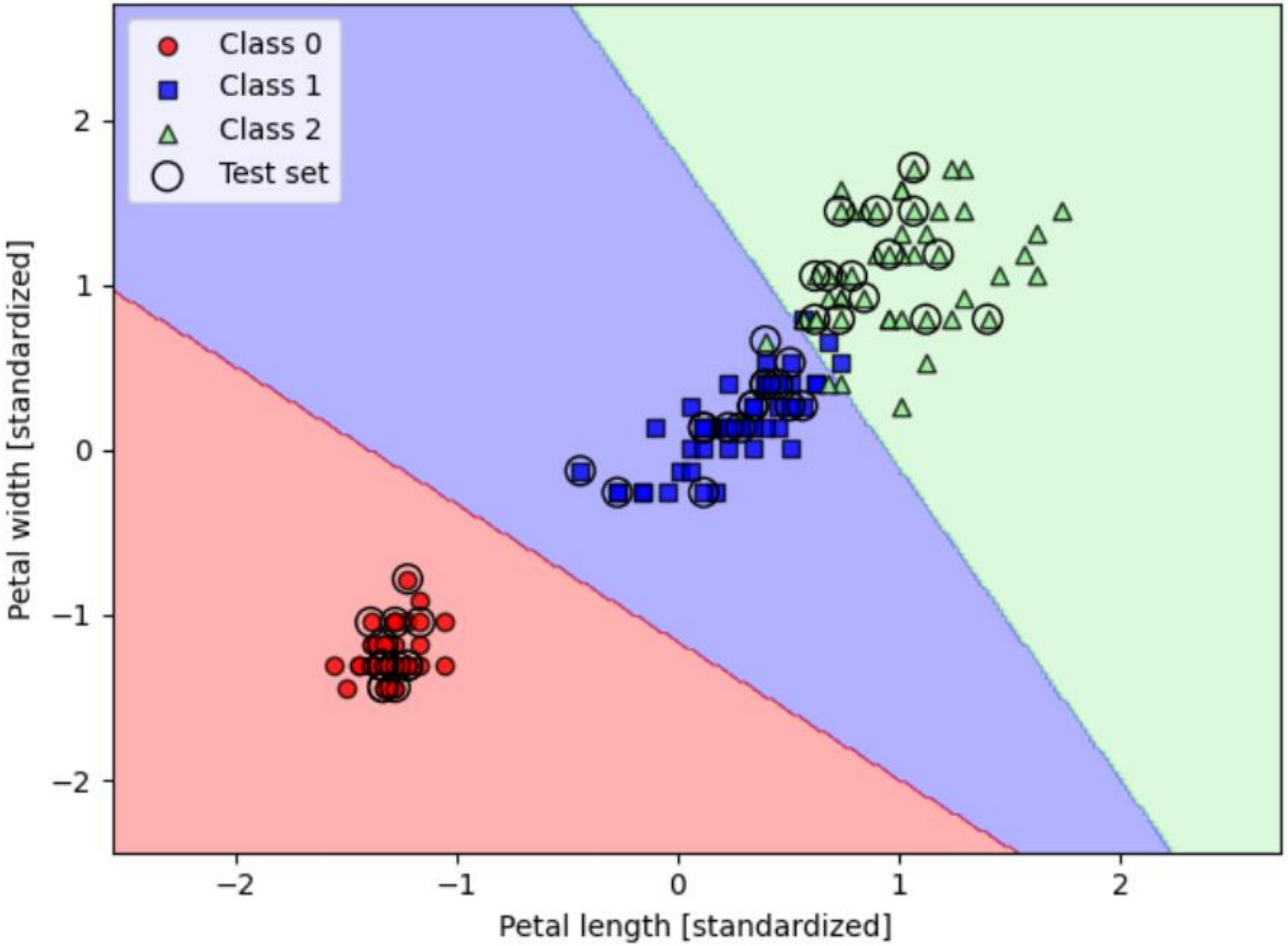
$$\Delta \boldsymbol{w} = -\eta \nabla_{\boldsymbol{w}} L(\boldsymbol{w}, b), \quad \Delta b = -\eta \nabla_b L(\boldsymbol{w}, b)$$

gradients

Binary classification of the iris dataset



Multi-class classification



See section 2.3 (p. 24) of [2]

References

1. S. Raschka, Y. Liu and V. Mirjalili. Machine Learning with PyTorch and Scikit-Learn. Packt Publishing. 2022. https://sebastianraschka.com/blog/2022/ml-pytorch-book.html

2. A. Ng. CS229 Lecture Notes. Stanford University. 2023. https://cs229.stanford.edu/main_notes.pdf

3. T. Broderick. Introduction to Machine Learning course (6.036). Massachusetts Institute of Technology. 2020. https://tamarabroderick.com/ml.html

Selected Video from youtube

- **Stanford CS109 Probability for Computer Scientists I Logistic Regression I 2022 I Lecture 24** **https://youtu.be/ILqZWvDWKEc?si=p0CL8fiRuzdwJ42_**
- **Stanford CS229 I Weighted Least Squares, Logistic regression, Newton's Method I 2022 I Lecture 3** **https://youtu.be/k_pDh_68K6c?si=A5PRwXa0w0L7zwTf**